

Exp no: 13 Determination and plotting of intrinsic impedance and phase velocity of various medium

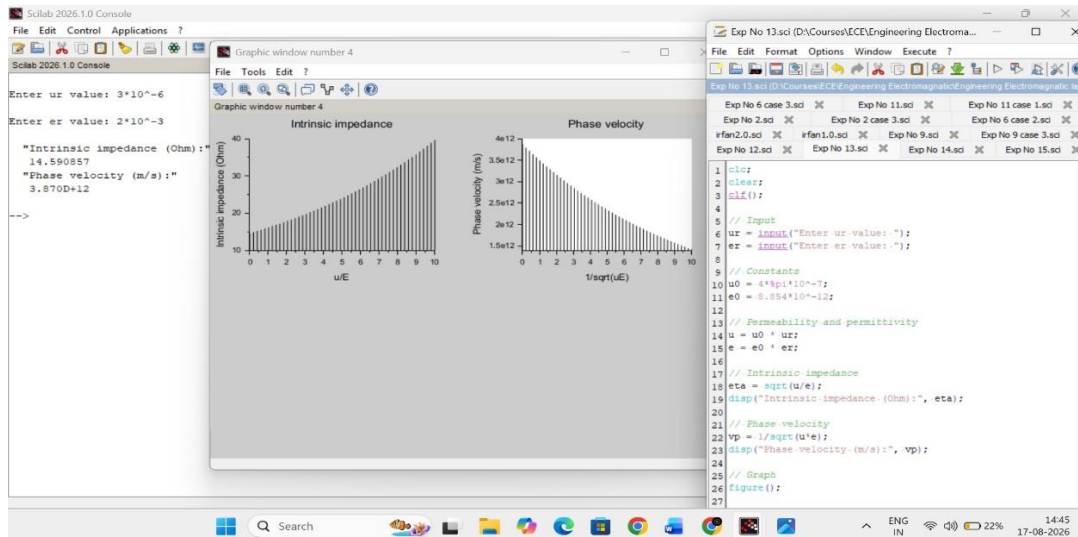
Date:

Aim:

To determine and plot intrinsic impedance and phase velocity of various medium.

Software required:

- SCILAB version 6.0.1



Exr No. 13

Determination of Intrinsic Impedance and
Phase velocity

$$\mu_r = 3 \times 10^{-6}$$

$$\epsilon_r = 2 \times 10^{-3}$$

Intrinsic Impedance

$$\eta = \sqrt{\frac{\mu}{\epsilon}}$$

where

$$\mu = \mu_0 \mu_r$$

$$\epsilon = \epsilon_0 \epsilon_r$$

$$\eta = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}}$$

$$= \sqrt{\frac{(4\pi \times 10^{-7}) (3 \times 10^{-6})}{8.854 \times 10^{-12} (2 \times 10^{-3})}}$$

$$= \sqrt{\frac{3.768 \times 10^{-12}}{8.854 \times 10^{-12}}}$$

$$\eta = 176.629 \Omega$$

$$\eta = \sqrt{21289}$$

$$\boxed{\eta = 14.59}$$

Phase velocity

$$V_p = \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{(\mu_0 \mu_r)(\epsilon_0 \epsilon_r)}}$$

$$= \frac{1}{\sqrt{(3.796 \times 10^{-12})(1.7708 \times 10^{-4})}}$$

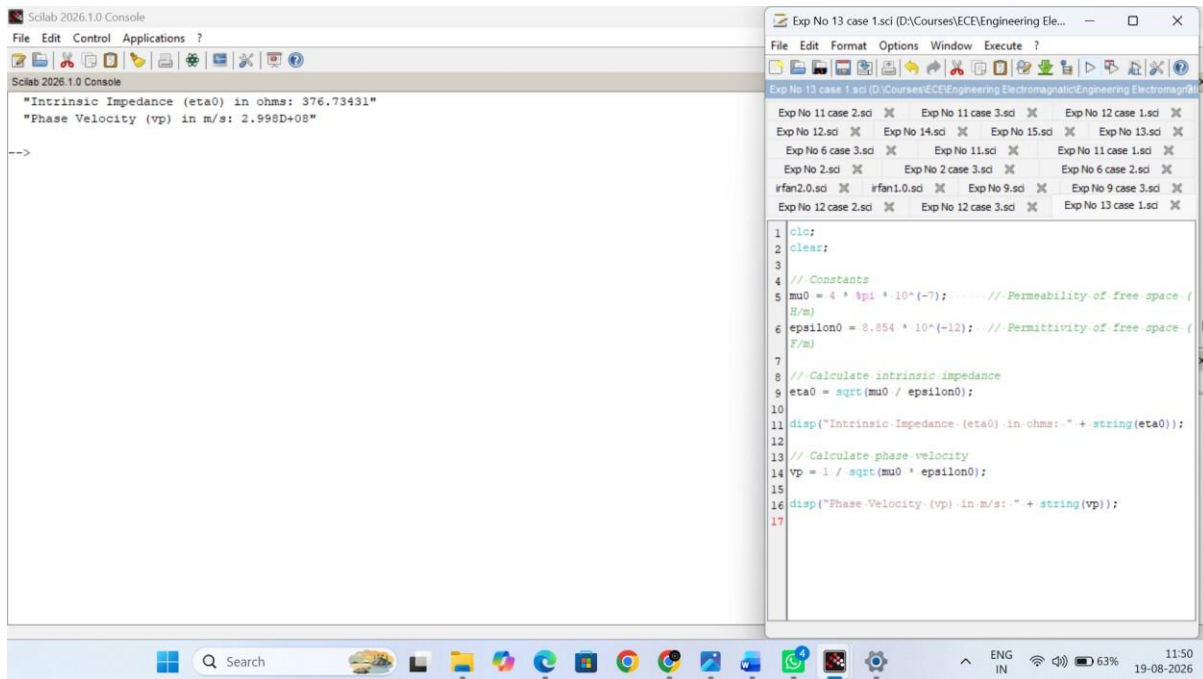
$$V_p = 3.87 \times 10^8 \text{ m/s}$$

$$\boxed{V_p = 3.87 \times 10^8 \text{ m/s}}$$

Result:

Thus, the determination and plotting of intrinsic and phase velocity is performed and successfully verified.

Case 1: To determine the intrinsic impedance and phase velocity of electromagnetic waves in free space (vacuum).



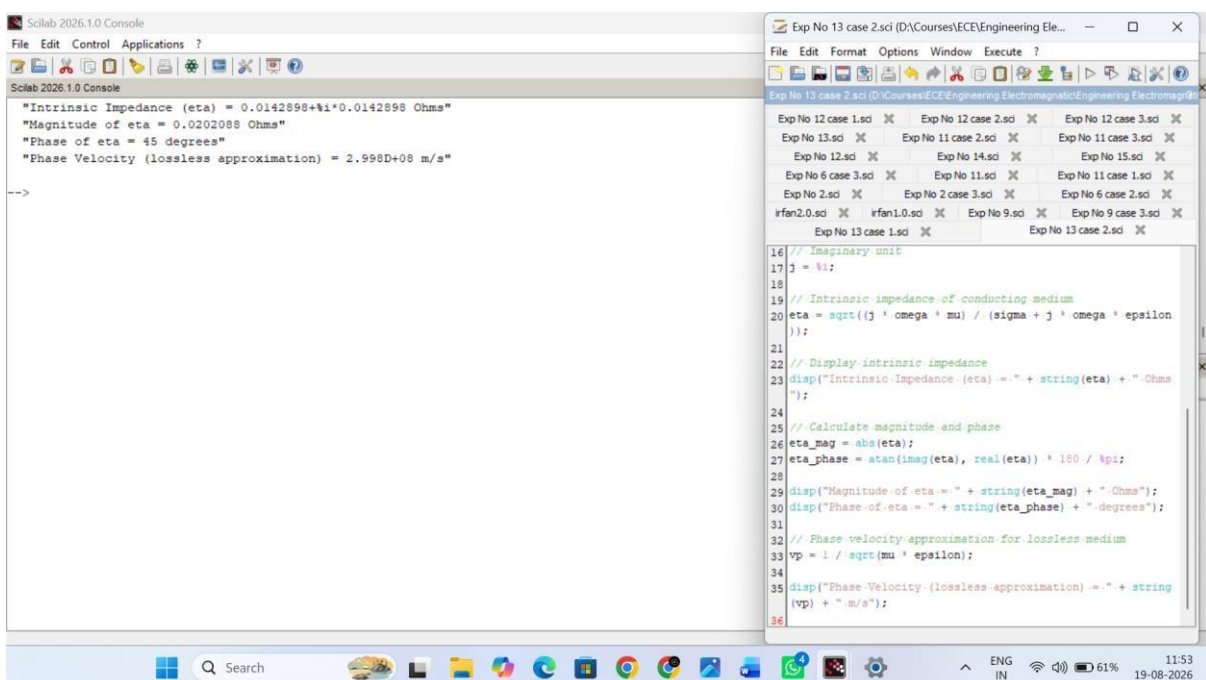
The screenshot shows the Scilab 2026.1.0 interface. The console on the left displays the output of the script: "Intrinsic Impedance (eta0) in ohms: 376.73431" and "Phase Velocity (vp) in m/s: 2.998D+08". The editor on the right shows the following code:

```
1 clc;
2 clear;
3
4 // Constants
5 mu0 = 4 * %pi * 10^(-7); // Permeability of free space (H/m)
6 epsilon0 = 8.854 * 10^(-12); // Permittivity of free space (F/m)
7
8 // Calculate intrinsic impedance
9 eta0 = sqrt(mu0 / epsilon0);
10
11 disp("Intrinsic Impedance (eta0) in ohms: " + string(eta0));
12
13 // Calculate phase velocity
14 vp = 1 / sqrt(mu0 * epsilon0);
15
16 disp("Phase Velocity (vp) in m/s: " + string(vp));
17
```

Result:

Thus the determination and plotting of intrinsic and phase velocity is performed and successfully verified in free space.

Case 2: To determine the intrinsic impedance and phase velocity for a wave propagating through a conductive medium like copper.



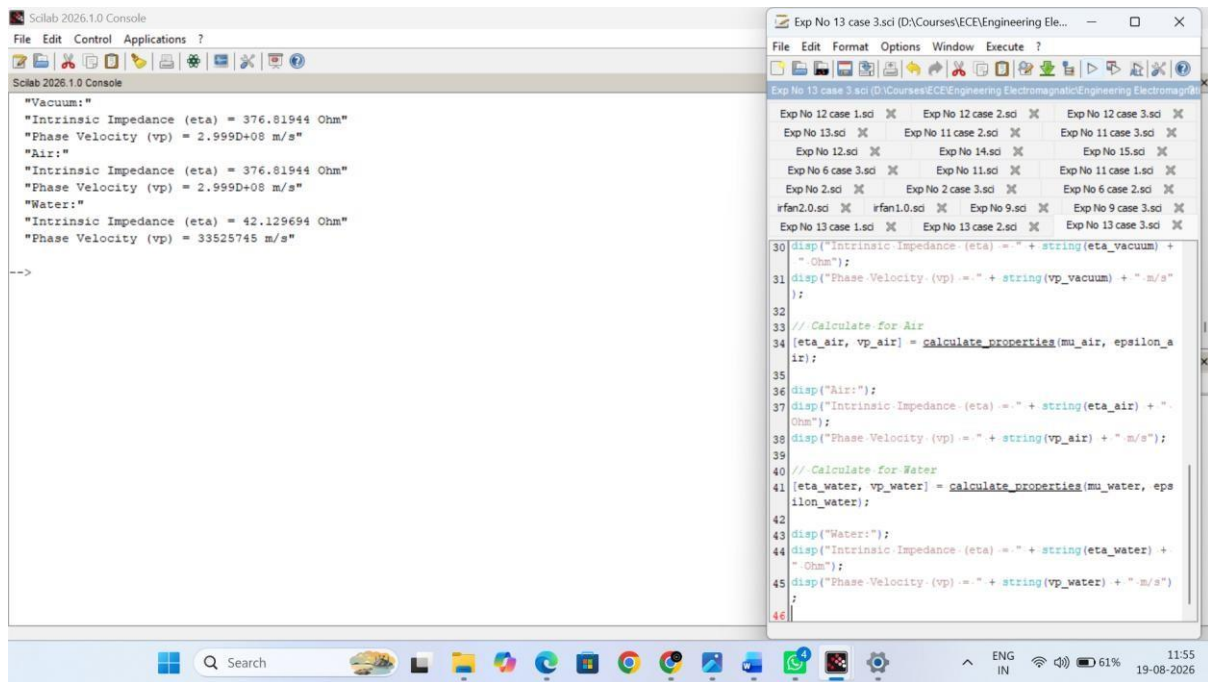
The screenshot shows the Scilab 2026.1.0 interface. The console on the left displays the output of the script: "Intrinsic Impedance (eta) = 0.0142898+1i*0.0142898 Ohms", "Magnitude of eta = 0.0202088 Ohms", "Phase of eta = 45 degrees", and "Phase Velocity (lossless approximation) = 2.998D+08 m/s". The editor on the right shows the following code:

```
16 // Imaginary unit
17 j = %i;
18
19 // Intrinsic impedance of conducting medium
20 eta = sqrt((j * omega * mu) / (sigma + j * omega * epsilon));
21
22 // Display intrinsic impedance
23 disp("Intrinsic Impedance (eta) = " + string(eta) + " Ohms");
24
25 // Calculate magnitude and phase
26 eta_mag = abs(eta);
27 eta_phase = atan(imag(eta), real(eta)) * 180 / %pi;
28
29 disp("Magnitude of eta = " + string(eta_mag) + " Ohms");
30 disp("Phase of eta = " + string(eta_phase) + " degrees");
31
32 // Phase velocity approximation for lossless medium
33 vp = 1 / sqrt(mu * epsilon);
34
35 disp("Phase Velocity (lossless approximation) = " + string(vp) + " m/s");
36
```

Result:

Thus, the determination and plotting of intrinsic and phase velocity is performed and successfully verified in copper.

Case 3: To determine and compare the intrinsic impedance and phase velocity of electromagnetic waves propagating through different media: Vacuum, Air, and Water.



The screenshot displays the Scilab 2026.1.0 environment. The console window on the left shows the output of the calculations for Vacuum, Air, and Water. The editor window on the right shows the corresponding Scilab code.

Console Output:

```
"Vacuum:"  
"Intrinsic Impedance (eta) = 376.81944 Ohm"  
"Phase Velocity (vp) = 2.999D+08 m/s"  
"Air:"  
"Intrinsic Impedance (eta) = 376.81944 Ohm"  
"Phase Velocity (vp) = 2.999D+08 m/s"  
"Water:"  
"Intrinsic Impedance (eta) = 42.129694 Ohm"  
"Phase Velocity (vp) = 33525745 m/s"
```

Editor Code (Exp No 13 case 3.sci):

```
30 disp("Intrinsic Impedance (eta) = " + string(eta_vacuum) +  
    " Ohm");  
31 disp("Phase Velocity (vp) = " + string(vp_vacuum) + " m/s"  
    );  
32  
33 // Calculate for Air  
34 [eta_air, vp_air] = calculate_properties(mu_air, epsilon_a  
    ir);  
35  
36 disp("Air:");  
37 disp("Intrinsic Impedance (eta) = " + string(eta_air) + "  
    Ohm");  
38 disp("Phase Velocity (vp) = " + string(vp_air) + " m/s");  
39  
40 // Calculate for Water  
41 [eta_water, vp_water] = calculate_properties(mu_water, eps  
    ilon_water);  
42  
43 disp("Water:");  
44 disp("Intrinsic Impedance (eta) = " + string(eta_water) +  
    " Ohm");  
45 disp("Phase Velocity (vp) = " + string(vp_water) + " m/s")  
    ;  
46
```

Result: Thus, the determination and plotting of intrinsic and phase velocity is performed and successfully verified in copper.

